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10 / 10 submitted

Q1. Quadratic equation

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>
#include <math.h>

int main() {
    double a, b, c, d, r1, r2;

    scanf("%lf %lf %lf", &a, &b, &c);

    d = b * b - 4 * a * c;

    if (d >= 0) {
        r1 = (-b + sqrt(d)) / (2 * a);
        r2 = (-b - sqrt(d)) / (2 * a);
        printf("%.2lf %.2lf", r1, r2);
    }

    return 0;
}
```

INPUT 1 -5 6

OUTPUT 3.00 2.00

Q2. Prime numbers

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int n, i, j, prime;

    scanf("%d", &n);

    if (n < 2) {
        printf("No prime numbers");
        return 0;
    }

    for (i = 2; i ≤ n; i++) {
        prime = 1;

        for (j = 2; j * j ≤ i; j++) {
            if (i % j == 0) {
                prime = 0;
                break;
            }
        }

        if (prime)
            printf("%d ", i);
    }

    return 0;
}
```

INPUT

20

OUTPUT

2 3 5 7 11 13 17 19

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Q4. Pyramid

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int n, i, j, space;

    if (scanf("%d", &n) != 1) {
        return 0;
    }

    for (i = 1; i ≤ n; i++) {
        // 1. Print leading spaces to center the pyramid
        for (space = 1; space ≤ n - i; space++) {
            printf(" ");
        }

        // 2. Print the ascending numbers (e.g., 1 2 3)
        for (j = 1; j ≤ i; j++) {
            printf("%d", j);
        }

        // 3. Print the descending numbers (e.g., 2 1)
        for (j = i - 1; j ≥ 1; j--) {
            printf("%d", j);
        }

        // Move to the next line
        printf("\n");
    }

    return 0;
}
```

INPUT

1

OUTPUT

1

Q5. Voting

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>

int main()
{
    int age;
    scanf("%d", &age);

    if (age ≥ 18)
        printf("Eligible");
    else
        printf("Not eligible");

    return 0;
}
```

INPUT

10

OUTPUT

Not eligible

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Q6. Calculator

Score: 0.00 TC: 0/3 passed

CODE

```
#include <stdio.h>

int main() {
    int a, b;
    char op;

    // Input format: 8*7 or 15/0 or 7+3
    scanf("%d%c%d", &a, &op, &b);

    switch(op) {
        case '+':
            printf("Add result = %d", a + b);
            break;

        case '-':
            printf("Subtraction result = %d", a - b);
            break;

        case '*':
            printf("Multiplication result = %d", a * b);
            break;

        case '/':
            if (b == 0)
                printf("Error: Division by zero");
            else
                printf("Division result = %d", a / b);
            break;

        case '%':
            if (b == 0)
                printf("Error: Division by zero");
            else
                printf("Modulo result = %d", a % b);
            break;

        default:
            printf("Invalid operator");
    }

    return 0;
}
```

INPUT 15 / 0

OUTPUT Invalid operator

Q3. Palindrome

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i, len, flag = 1;

    if (scanf("%s", str) != 1) {
        return 0;
    }

    len = strlen(str);

    for (i = 0; i < len / 2; i++) {
        if (str[i] != str[len - i - 1]) {
            flag = 0;
            break;
        }
    }

    if (flag)
        printf("Palindrome\n");
    else
        printf("Not a Palindrome\n");

    return 0;
}
```

INPUT madam

OUTPUT Palindrome

Q7. Month counting

Score: 0.00 TC: 0/2 passed

CODE

```
#include <stdio.h>

int main() {
    int month;

    scanf("%d", &month);

    switch(month) {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            printf("31");
            break;

        case 4:
        case 6:
        case 9:
        case 11:
            printf("30");
            break;

        case 2:
            printf("28");
            break;

        default:
            printf("Invalid month");
    }

    return 0;
}
```

INPUT

2

OUTPUT

28

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Q8. Positive or negative

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>

int main() {
    int n;

    scanf("%d", &n);

    if (n > 0)
        printf("Positive");
    else if (n < 0)
        printf("Negative");
    else
        printf("Zero");

    return 0;
}
```

INPUT -5

OUTPUT Negative

Q9. Multiplication Table

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>

int main() {
    int n, i;

    scanf("%d", &n);

    for(i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }

    return 0;
}
```

INPUT 5

OUTPUT
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

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Q10. Factorial

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int n, i;
    long long fact = 1;

    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        fact *= i;
    }

    printf("%lld", fact);

    return 0;
}
```

INPUT 5

OUTPUT 120